

BORA paper-data: figure/table review

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Table 1: Evaluation datasets: MAL (malware scanning), DNA (genomic variants), and DLP (e-mail DLP). The *Rule shape* column summarizes the form and variety of the regexes.

	Document corpus				Regex set			
	Corpus	Docs	Size	Per-doc (min/med/max)	Ruleset (ver.)	Rules	Size	Rule shape
MAL	CentOS 7	1,209	765 MB	66 KB / 130 KB / 33 MB	ClamAV 0.103.11	38,875	11.6 MB	literal rich; 160,854 leaf subsigs (561 PCRE); logical AND/OR combinations; frequent bounded gaps $\{n\}$; mostly unanchored, implicit $.*\dots*$ shape
DNA	chr17 (GRCh38)	1	83 MB	— (single)	NCBI ClinVar (chr17)	27,500	1.4 MB	anchored $\wedge.\{N\}\text{literal}.*$; len 44–8,738 chars (avg 51)
DLP	Enron Email	509,610	1.38 GB	398 B / 1.5 KB / 1.7 MB	MS-DLP (SIT)	136	7 KB	keyword-proximity: $(kw_1 \dots kw_n)\{0,300\}\text{re} re.\{0,300\}(kw_1 \dots kw_n)$

Dataset	Pairs	CP	SDE	DFA
MAL	4.70×10^7	99.8803%	0.1144%	0.0053%
DNA	2.75×10^4	99.9382%	0.0618%	0.0000%
DLP	4.98×10^9	99.9980%	0.0020%	0.0000%

Table 2: Tier-discharge fractions per dataset.

Size (B)	Files	CP	SDE	DFA
$2^{16}\text{--}2^{19}$	954	99.9140%	0.0808%	0.0052%
$2^{19}\text{--}2^{21}$	220	99.7566%	0.2380%	0.0054%
$2^{21}\text{--}2^{23}$	21	99.8119%	0.1809%	0.0072%
$2^{23}\text{--}2^{25}$	14	99.6333%	0.3577%	0.0090%

Table 3: Tier-discharge fractions by file size (Mal).

Size (B)	Files	CP	SDE	DFA
$2^7\text{--}2^{11}$	326235	99.9992%	0.0008%	0.0000%
$2^{11}\text{--}2^{14}$	172337	99.9965%	0.0035%	0.0000%
$2^{14}\text{--}2^{17}$	6130	99.9793%	0.0207%	0.0000%
$2^{17}\text{--}2^{20}$	152	99.9384%	0.0616%	0.0000%

Table 4: Same, over Dlp.

Component	Purpose	MAL		DNA		DLP	
		M	%	M	%	M	%
range table [†]	range proof e.g. for pattern distance	67.1	86.5	134.2	98.5	33.6	99.5
AC-DFA / CP	transition rules	3.1	4.0	1.0	0.7	0.1	0.2
SDE	subsig step pattern distance constraints	7.4	9.5	1.0	0.7	0.1	0.2
DFA (3/0/0)	transition rules	0.0	0.0	0.0	0.0	0.0	0.0
sig-DB	DNF combinations / tri-logic truth table	0.0	0.0	0.0	0.0	0.0	0.0
TOTAL populated		77.6	28.9	136.2	50.7	33.7	12.6
#signatures		1,944		1,375		494	

Table 5: Lookup-table composition across datasets. Column M is millions of entries. Each % is of that dataset’s populated entries; TOTAL % is of the 2^{28} capacity. [†]range table = $2^{\text{range}2.\text{bit}}$ (26 / 27 / 25 for MAL/DNA/DLP), sized by the maximum document offset. The $(\cdot / \cdot / \cdot)$ after *DFA* is the number of DFAs folded in for rules that frequently fail SDE, per dataset. BORA re-writes $(\text{kw1} | \dots \text{kwN}) \cdot \{0, N\} \text{pat}$ in DLP to disjunction of n rules for improved SDE accuracy, thus expanding from 136 to 9861.

Dataset	Circ	input		R1CS / input byte: raw (adjusted)				corpus
		(B)	R1CS	c_1 (CP)	c_2 (SDE)	c_3 (DFA)	c_4 (frwk)	
MAL	0	3,968	380,454	9.3 (29.0)	18.6 (57.1)	4.5 (4.8)	63.5 (5.0)	3.6%
	1	3,968	435,331	10.4 (27.9)	27.0 (72.0)	4.5 (4.7)	67.9 (5.1)	96.4%
DNA	0	7,936	885,871	28.7 (63.9)	22.6 (45.4)	0.0 (0.0)	60.4 (2.4)	100.0%
DLP	0	1,984	162,236	5.7 (12.3)	23.8 (51.8)	0.0 (0.0)	52.2 (17.7)	100.0%
	1	1,984	163,880	5.7 (12.2)	24.4 (52.7)	0.0 (0.0)	52.5 (17.7)	0.0%

Table 6: Per-circuit cost profile. Each dataset is discharged by a small set of circuits. *input* is the bytes processed per folding step. Each $c_i \in (c_1, c_2, c_3, c_4)$ is the per-input-byte R1CS constraints of the four components of a circuit. Each c_i is shown as *raw (adjusted)*: the *adjusted* value moves the *measured* log-up query cost in c_4 and charges it to each tier in proportion to its data-column size, so $c_1/c_2/c_3$ reflect the all-in per-byte cost and c_4 retains only folding overhead and the processing of Logup share of lookup table.

Dataset	Corpus	Jobs	Steps	Stage 1: Main Folding			Total / Wall (jobs)	Stage 2
				Sel.	Commit wit.	Fold		Compression
MAL	829.2 KB	2	214	35.2 s	1.3 min	4.7 min	6.6 min / 3.3 min (2)	58.4 s
DNA	813.8 KB	1	105	19.8 s	34.8 s	3.6 min	4.5 min (1)	57.2 s
DLP	350.7 KB	2	181	2.5 min	1.9 min	9.3 min	13.8 min / 6.9 min (2)	1.3 min
Per batch proof (one per job, constant): size 1,152 B, verification 25 ms.								

Table 7: End-to-end prover performance. *Corpus* is the bytes actually folded: DNA packs each base of its chr17 sequence into a nibble (half a byte), halving it to the 813.8 KB shown, while the 829.2 KB MAL results from padding each chunk to 128 KB. *Stage 1* (main folding) is the corpus-scaling cost: circuit selection + commit fixed witness + folding. *Total / Wall (jobs)* reports the summed CPU-time over the parallel jobs and the wall-clock time; for balanced jobs the wall-clock is $\approx$ Total/Jobs. *Stage 2: Compression* computes the main-circuit SNARK proof, the QA-NIZK proof, the cycle-pairing fold, and the resulting batch proof. It is a fixed per-job cost, and all timings exclude the one-time key setup. The bottom row shows the size and verification cost of one batch proof (mainly the two Groth16 SNARK proofs); this per-job batch proof is constant size and independent of corpus and rule-set size. A run emits one batch proof per job, as jobs split a corpus to save wall time.

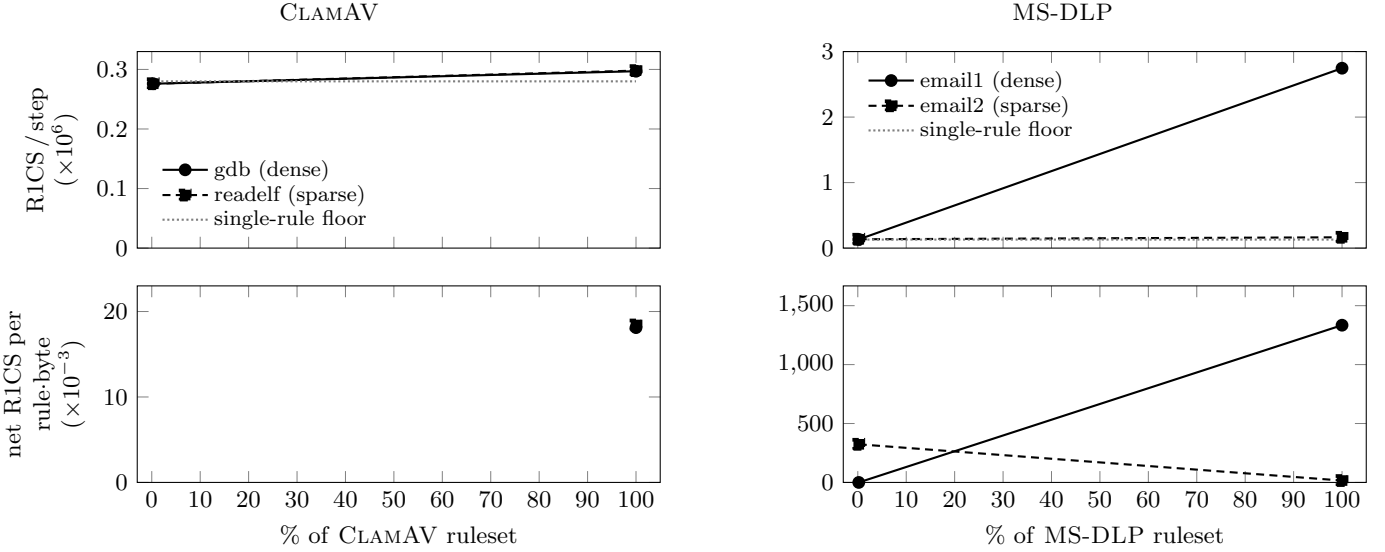


Figure 1: Ruleset-size scalability (input fixed), per folding step, on CLAMAV (left) and MS-DLP (right). *Top*: absolute circuit size with the single-rule framework floor (dotted; 0.28 M R1CS for CLAMAV, 0.13 M for MS-DLP); growing the ruleset to the full 300 (CLAMAV) / 987 (MS-DLP) rules grows the circuit far slower than the rule count. *Bottom*: net circuit cost *per rule per input byte*—cumulative R1CS above the floor divided by cumulative rule count times per-step input length—falling from 18.11×10^{-3} to 18.11×10^{-3} (**gdb**) and 18.42×10^{-3} to 18.42×10^{-3} (**readelf**) R1CS per rule-byte (CLAMAV) and 1334.28×10^{-3} to 1334.28×10^{-3} (**email1**) and 323.59×10^{-3} to 16.03×10^{-3} (**email2**) R1CS per rule-byte (MS-DLP) at the full set. For the same MS-DLP ruleset, ZOMBIE emits 2,621,440 R1CS in total; dividing by the 2000-byte input and the same 987 rules gives ~ 1.33 R1CS per rule-byte, the same normalization applied to BORA.

Table 8: Reef per-bucket cost vs. BORA on chr17 $\times$ NCBI (27,500 regexes). Both systems are compared on the *net* step-folding cost only, excluding commitment, setup, and IVC proof compression, for a fair comparison. For Reef, net = witness generation + Nova prove + SAFA solve (**fa_solver**); it excludes the fixed floor (the 4.3 GB commitment load, SNARK setup, consistency proof, and related one-time costs). For BORA, net = the Phase 1 circuit selection and non-deterministic advice generation, and Phase 2 main-folding step cost, excluding the Groth16 IVC proof compression as well as setup and commitment. Estimate1 uses each bucket’s measured per-run mean times its population count; Estimate2 uses the minimum measured per-run mean (proj_512k) times bucket count as a counterfactual floor.

Bucket	Count	Per-run net cost (s)	Estimate1 (s)	Estimate2 (s)
non_projectable	24,434	1036.18 ± 0.00	2.532×10^7	1.657×10^5
proj_512k	58	6.78 ± 0.00	3.932×10^2	3.932×10^2
proj_2M	132	13.54 ± 0.00	1.787×10^3	8.950×10^2
proj_4M	1,194	25.76 ± 0.00	3.076×10^4	8.095×10^3
proj_8M	869	51.70 ± 0.00	4.493×10^4	5.892×10^3
proj_16M	813	88.41 ± 0.00	7.188×10^4	5.512×10^3
Reef TOTAL	27,500	—	2.547×10^7 (294.77 d)	1.865×10^5 (2.16 d)
BORA	27,500	— (single batched run)	2.707×10^2 (4.5 min) — $\approx 94088\times$ faster	2.707×10^2 (4.5 min) — $\approx 688.82\times$ faster

Table 9: Zombie prover cost on the MS-DLP international SIT rules (Spartan NIZK, two-pattern proximity circuit), totalled over all 4 regex instances (the 4 SIT rules of Table 1, expanded into their comb variants) at each scanned document length L_T . The total regex size (sum of pattern + keyword bytes over all instances, each sub-pattern counted once) is fixed at 7,443 B. The last row derives the unit cost $u = t_{\text{prove}} / (L_T \cdot \sum s)$ in seconds per (input-byte $\times$ regex-byte). Rows marked † have no real measurement at that length in this log (e.g. a dry-run sweep with a smaller **VEC_SIZE**) – all four totals are linearly interpolated from the 1,000 B row, under a linear-growth assumption matching Zombie’s per-byte circuit complexity; treat them as illustrative, not measured.

L_T (B)	Total constraints	Prove (s)	Verify (s)	Proof (MB)
700	1,179,648	9.2	4.2	0.11
800	1,310,720	25.4	15.0	0.14
1,000	1,310,720	13.5	7.6	0.16
2,000 †	2,621,440	27.1	15.3	0.31
Unit $u = t / (L_T \cdot \sum s)$ at $L_T = 700/800/1,000/2,000$: $1.77 \times 10^{-6} / 4.26 \times 10^{-6} / 1.82 \times 10^{-6} / 1.82 \times 10^{-6}$ s B $^{-2}$				

Table 10: Projected Zombie cost (unit $\bar{u} = 1.82 \times 10^{-6} \text{ sB}^{-2}$, taken at $L_T = 2,000$) vs. BORA’s net main-circuit folding cost. Est. Zombie = $\bar{u} \cdot (\text{corpus}) \cdot (\text{regex set})$, a full document $\times$ regex cross product. Regex-set size is on-disk rule-file bytes for MAL/DNA; for DLP it is the pattern+keyword bytes fed to the circuit (the on-disk `.regex` doubles each sub-pattern for the bidirectional proximity form). $\bar{u}$ itself is linearly interpolated (see Table 9’s $\dagger$ note): this log has no real measurement at $L_T = 2,000$, e.g. a dry-run sweep. The MAL (ClamAV) BORA cost 0.11 hr is the total over its 2 parallel jobs; the wall-clock time is 0.09 hr (the slowest job). The DLP (Enron) BORA cost 0.23 hr is the total over its 2 parallel jobs; the wall-clock time is 0.12 hr (the slowest job).

Dataset	Corpus	Regex set	Est. Zombie	BORA (net)	Speedup
MAL (ClamAV)	765.5 MB	11.6 MB	1.62×10^{10} (513.6 yr)	3.95×10^2 (0.11 hr)	$\approx 40,987,246\times$
DNA (chr17)	83.3 MB	1.4 MB	2.14×10^8 (6.8 yr)	2.71×10^2 (0.08 hr)	$\approx 790,573\times$
DLP (Enron)	1.42 GB	7.4 KB	1.92×10^7 (222.6 d)	8.26×10^2 (0.23 hr)	$\approx 23,297\times$

Table 11: End-to-end prover cost of Zombie [?], Reef [?], and BORA on the three datasets of Table 1; the last two columns give BORA’s speedup over each. Zombie costs are projected from the unit cost $u = 1.82 \times 10^{-6} \text{ sB}^{-2}$ (Table 9). For DNA, Reef’s as-implemented cost is shown with its optimistic floor in parentheses. Values marked * are projected from Reef’s cheapest measured per-run cost on DNA (6.78 s, the `proj_512k` bucket): the DNA floor as that cost per signature, and Reef MAL/DLP as that cost per signature per corpus byte (Table 8). MAL BORA is the total net cost summed over 2 parallel jobs; the wall-clock time is 0.09 hr (the slowest job). DLP BORA is the total net cost summed over 2 parallel jobs; the wall-clock time is 0.12 hr (the slowest job). Cells shown as “??” have no audited measurement yet.

Dataset	Zombie	Reef	BORA	Speedup	
				vs Zombie	vs Reef
MAL	513.6 yr	6.4 Myr*	0.11 hr	41M $\times$	510B $\times$ *
DNA	6.8 yr	294.8 d (2.2 d*)	0.08 hr	790,573 $\times$	94,088 $\times$ (689 $\times$ *)
DLP	222.6 d	41,525 yr*	0.23 hr	23,297 $\times$	2B $\times$ *